

How to use SeedPhrasePoker

Two physical decks become a BIP-39 phrase

This file turns card pairs into a 12- or 24-word seed. The shuffle is the entropy. The page only looks the pair up, then computes the BIP-39 checksum. It does not pick cards for you and it does not talk to the network.

v2.0.2 is phrase-only. Write the words on paper or steel and restore them on a hardware wallet. Check the finished phrase on a second offline BIP-39 tool before any coins move.

What you need

- Two complete 52-card decks of the same size and finish. Leave the jokers out.
- Plastic casino-style cards shuffle cleanly and wear slowly. Copag and Cartamundi (or another plastic poker set of the same cut) are a good default. Two paper tourist decks of mixed brands stick, mark, and split in the hand.
- A pencil to mark twelve extra face cards with a clear X.
- The HTML file on an offline computer, already copied to disk. Wait for the green self-test banner.
- One printed pair booklet: seedphrase-poker-pairs-0based.pdf (numbers 0–2047, same as the page) or seedphrase-poker-pairs-1based.pdf (same pairs, numbers 1–2048). Pick one and stay with it.
- Pen and paper, or a steel plate, for the words. Do not photograph the finished phrase.

Use two matching plastic decks. Same size, same back, same finish. That keeps the shuffle honest and the cards readable after many draws.

Build the two decks

Blue deck — 32 cards. From deck 1 keep only Ace through 8 of every suit (♥ ♦ ♣ ♠). Set the rest aside. You will reuse the twelve face cards (J, Q, K of each suit) for the orange deck.

Orange deck — 64 cards. Take deck 2 in full. From the leftover face cards of deck 1, mark each Jack, Queen and King with a pencil X. A Jack of Hearts and a Jack of Hearts X are two different cards. Add those twelve marked cards to deck 2: $52 + 12 = 64$.

On the page the marked cards sit on their own row: JX, QX, KX. Suit order: hearts, diamonds, clubs, spades. Blue ranks: A 2 3 4 5 6 7 8. Orange ranks: A 2 3 4 5 6 7 8 9 10 J Q K, then the X row.

Shuffle

Shuffle each deck on its own. Seven or more riffle shuffles per deck is a reasonable minimum. A long face-down wash works too. Blue face-down on the left, orange face-down on the right.

Preferred rhythm: after every pair, return both cards and shuffle again. Repeats are allowed in BIP-39. Faster rhythm: one thorough shuffle, then deal without putting cards back. Do not do that for 23 entropy words from a 32-card blue deck — you run out of blue cards. Shuffle again if a pile is empty.

Draw and enter

Set 12 or 24 words on the page before the first pair. Length locks after that.

Draw the top blue card and the top orange card. On the page, tap that suit and rank in each deck, then Confirm entropy pair. Marked orange faces are JX, QX or KX on the lower row. Or look the same pair up in the booklet and write the word. Do this 11 times for a 12-word phrase, or 23 times for 24 words. Those are entropy words only.

The formula, if you want to check a line: word index = blue $\times$ 64 + orange. Always 0–2047.

The last word

A 12-word phrase is 128 bits of entropy plus a 4-bit SHA-256 checksum. Eleven pairs give 121 bits, so

the last pair keeps 7 bits and the page fills 4 checksum bits. A 24-word phrase is $256 + 8$; twenty-three pairs give 253 bits, so the last pair keeps 3 bits and the page fills 8.

Draw one extra pair and press Confirm checksum pair. The printed last word is usually not the raw pair word. That is required by BIP-39.

If you would rather not use this file for the last word, enter the 11 or 23 entropy words plus a candidate last word on an air-gapped hardware wallet. If it rejects the phrase, discard only that last pair and draw again. Chance of a hit: 1 in 16 for 12 words, 1 in 256 for 24 words. Do not invent the last word, and do not pick a favourite from the valid endings.

After the words

The finished phrase stays blurred until you press Show words. Write them down, then Wipe and close the tab. JavaScript cannot erase RAM. On Tails, shut the machine down when you are done.

This copy must show a green self-test on load. Compare the file on disk with seedphrase-poker-v2.0.2-SHA256SUMS.txt before you move it to an air-gapped machine.